

1 a) i. $-111011.11011 \times 2^{12} \rightarrow 26 \text{ bit IEEE 754.}$

$$-111011.11011 \times 2^{12} = -1.1101111011 \times 2^{17}$$

Fraction - 13 bits

1	12 bits	13 bits
1	10000 0010 000	11011 1101 1000

$$\text{Bias} = 2^{12-1} - 1 = 2047$$

$$\text{Biased exp} = 2047 + 17 = 2064$$

0011 0000 0010 0001 1011 1101 1000

3 0 2 1 B D 8

0x3021BD8

(ii) Fraction 17 bits. $\therefore \text{Exp} = 8 \text{ bits}$

$$\therefore \text{Bias} = 2^{8-1} - 1 = 127$$

$$\therefore \text{Biased exp} = 127 + 17 = 144 = 1001 0000$$

1	1001 0000	11011101 1000 0000
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0011 0010 0001 1011 1101 1000 0000

3 2 1 B D 8 0

$$\Rightarrow 0.082 = 0.0001010 = 1.01 \times 2^{-4}$$

$$0.108 = 0.0011001 = 1.1001 \times 2^{-3}$$

$$1.01 \times 2^{-4} \times 1.1001 \times 2^{-3}$$

$$\Rightarrow 1.111101 \times 2^{-7}$$

$$\text{actual exp} = -7$$

$$\therefore \text{Biased exp} = 127 - 7 = 120$$

$$1 \leq 120 \leq 127 \quad \therefore \text{Normal}$$

$$\textcircled{C} -6.55 = -110.10001 = -1.1010001 \times 2^2$$

$$15.21 = 1111.00110 = 1.11100110 \times 2^3$$

$$1.11100110 \times 2^3 - (-1.1010001 \times 2^2)$$

$$= 1.11100110 \times 2^3 + 1.1010001 \times 2^2$$

$$= 1.11100110 \times 2^3 + 0.11010001 \times 2^3$$

$$= 2^3 (1.11100110 + 0.11010001)$$

$$= 2^3 \times 10.1011011$$

$$= 1.01011011 \times 2^4$$

$$\begin{array}{r} 1.11100110 \\ 0.11010001 \\ \hline 10.1011011 \end{array}$$

$$110$$

$$1.2$$

$$1.2$$

$$10$$

2a(i) slli requires the third operand to be an immediate value. Operating with larger immediates we need to use lui.

ii) add x21, x22, x23

slli x21, x21, 5

lui x21, 0xh804

addi x21, x21, 0xh804

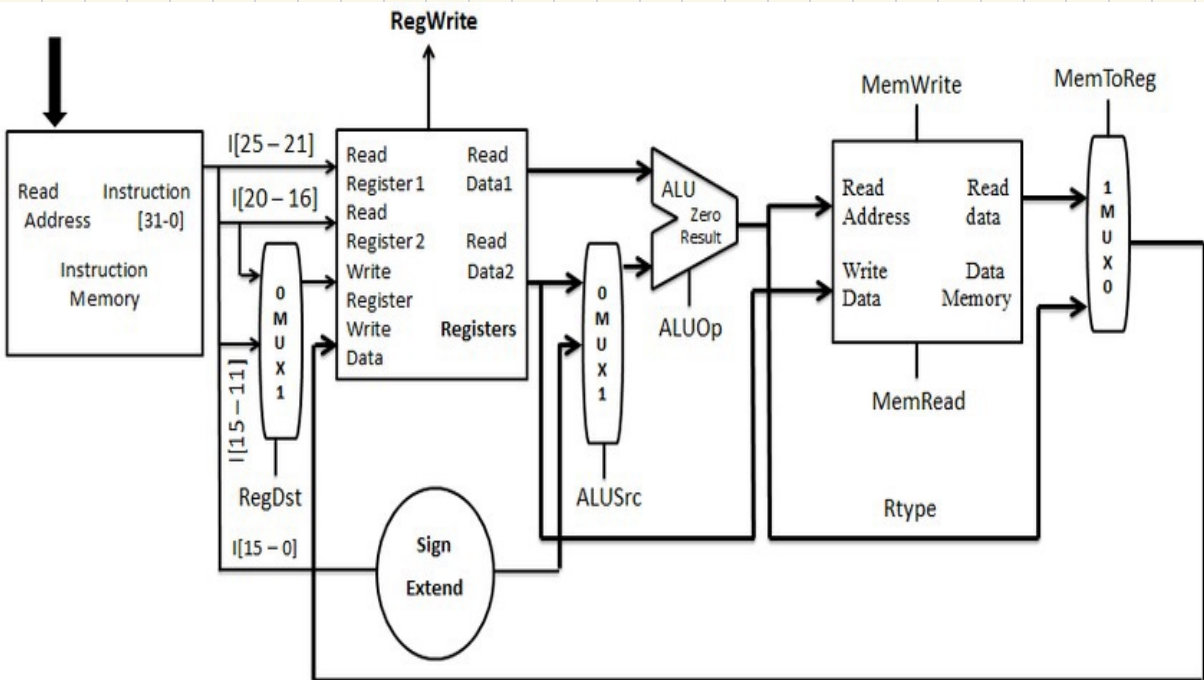
addi x22, x21, 15

b) 00CB1A93 = 0000 0000 1100 1011

0001 1010 1001 0011

0000 0000 1100 1011 001 10101 0010011

This is I type slli



3(a) RISC-V has different Immediate formats such as I type, S type, SB type. Different formats place immediate values at different positions. Such as I type [31:20], S type [31:25] & [11:7]. That's why the immediate generation unit must consider the entire instruction to determine the exact immediate value.

b i) 5 instructions, 5 cycles.

Avg CPI = 1. Since each instruction takes 1 cycle

ii) 4 data hazards. line 2, 3, 4, 5.

iii) IF ID EX MEM WB

x x x x x

x x x x x

IF ID EX MEM WB

x x x x x

x x x x x

IF ID EX MEM WB

x x x x x

x x x x x

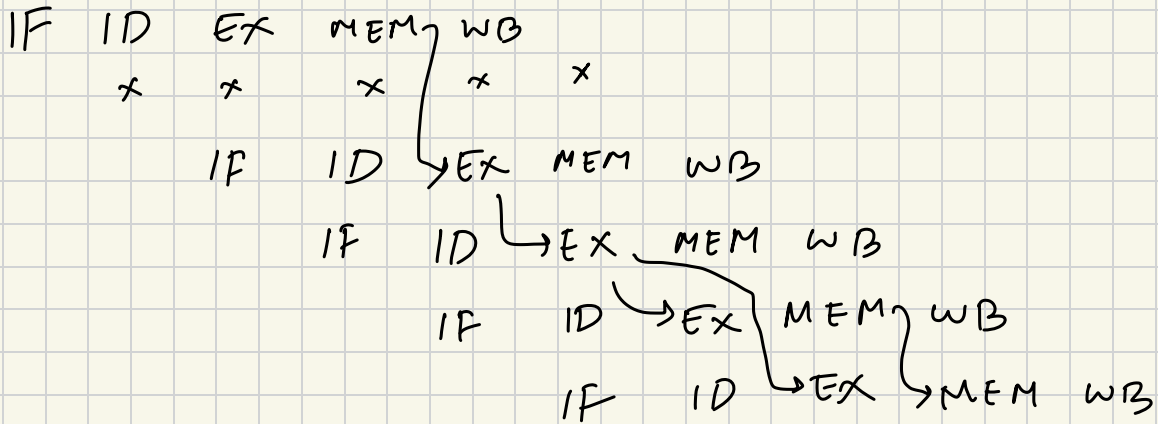
IF ID EX MEM WB

x x x x x

x x x x x

IF ID EX MEM WB

(iv) stall + forwarding:



(v) Total cycle = 10 using stall + forwarding

Total instruction = 5

$$\therefore CPI = \frac{10}{5} = 2$$

using only stall 17 cycles

$$\therefore CPI = \frac{17}{5} = 3.4$$